# SPM Notations

- MVP-L (Multi-view Process Modelling Language) represents a notation that offers a comprehensive set of modelling constructs.
- SPEM 2.0 (Software Process Engineering Meta-model) is also covered in the chapter as it has the potential to become a future process model notation standard.
- The Framework can be used to evaluate each of these two and any other type of SPM notations

# MVP-L

- 1st version (1989) at NASA Goddard Space Flight Center and University of Maryland by Dieter Rombach
- 2 subsequent versions at University of Kaiserslautern
- Supports process improvement by
  - Modelling
  - Planning
  - Enactment
  - Measurement and interpretation
  - Packaging
- Supports process modelling “in the large”
  - Clear interfaces
  - Level completeness



# MVP - Language Constructs

- Models (Types) for description of
  - Artefacts, e.g., documents (product_model)
  - Activities, e.g., review of a document (process_model)
  - Resources, e.g., people, tools (resource_model)
  - Quality aspects, e.g., effort prediction model (quality_model)
- Attributes
  - Refer to each model type
  - Attributes correspond to measures
  - Attribute values correspond to measurement data
- Models (types) are adapted to actual contexts through the instantiation to objects
- Executable objects are assembled in project plans (project_plan)

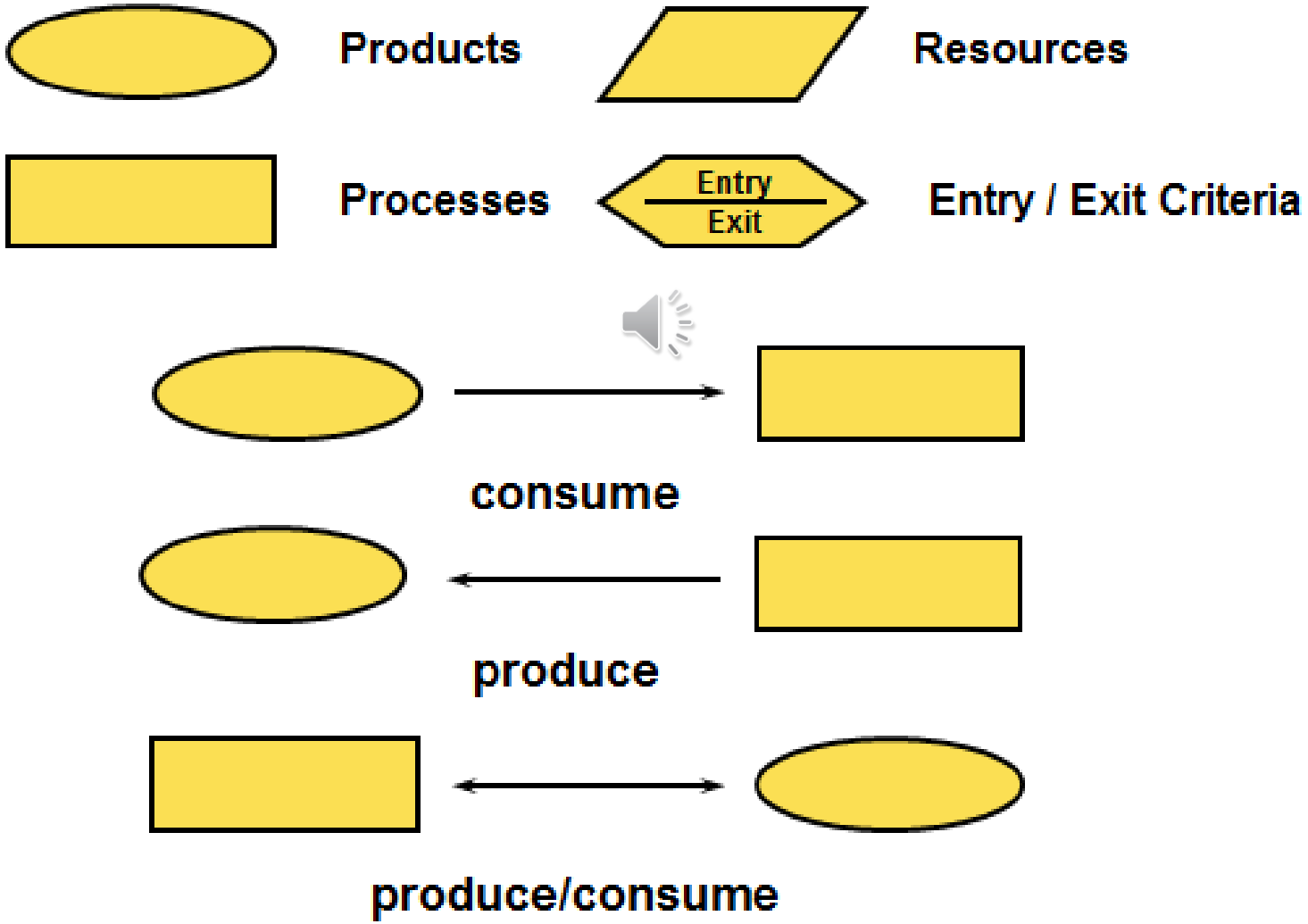
# Relationship between Objects

- Product flow
  - consume, produce, consume_produce
- Control flow
  - implicit via pre- and post-conditions and invariants (i.e., constraint-oriented description of control flows)
  - Constructs: entry_criteria, exit_criteria, and invariant
- Refinement / aggregation
- Information hiding
- Models are based on
  - Model-Interface (visible for other models) and
  - Model-Body (only visible internally)
- Models should be designed in such a way that potential changes are local

# Definition

- A project plan in MVP-L is a 13-tuple including the following components:
  - PC: Set of all processes
  - PD: Set of all products
  - RS: Set of all resources
  - ATT: Set of all attributes
  - BE: Set of all Boolean expressions
  - consume PD x PC
  - produce PC x PD
  - consume_produce PD x PC
  - attribute (PC U PD U RS) x ATT
  - entry PC x BE
  - exit PC x BE
  - resource PC x RS
  - S0 ATT → Value

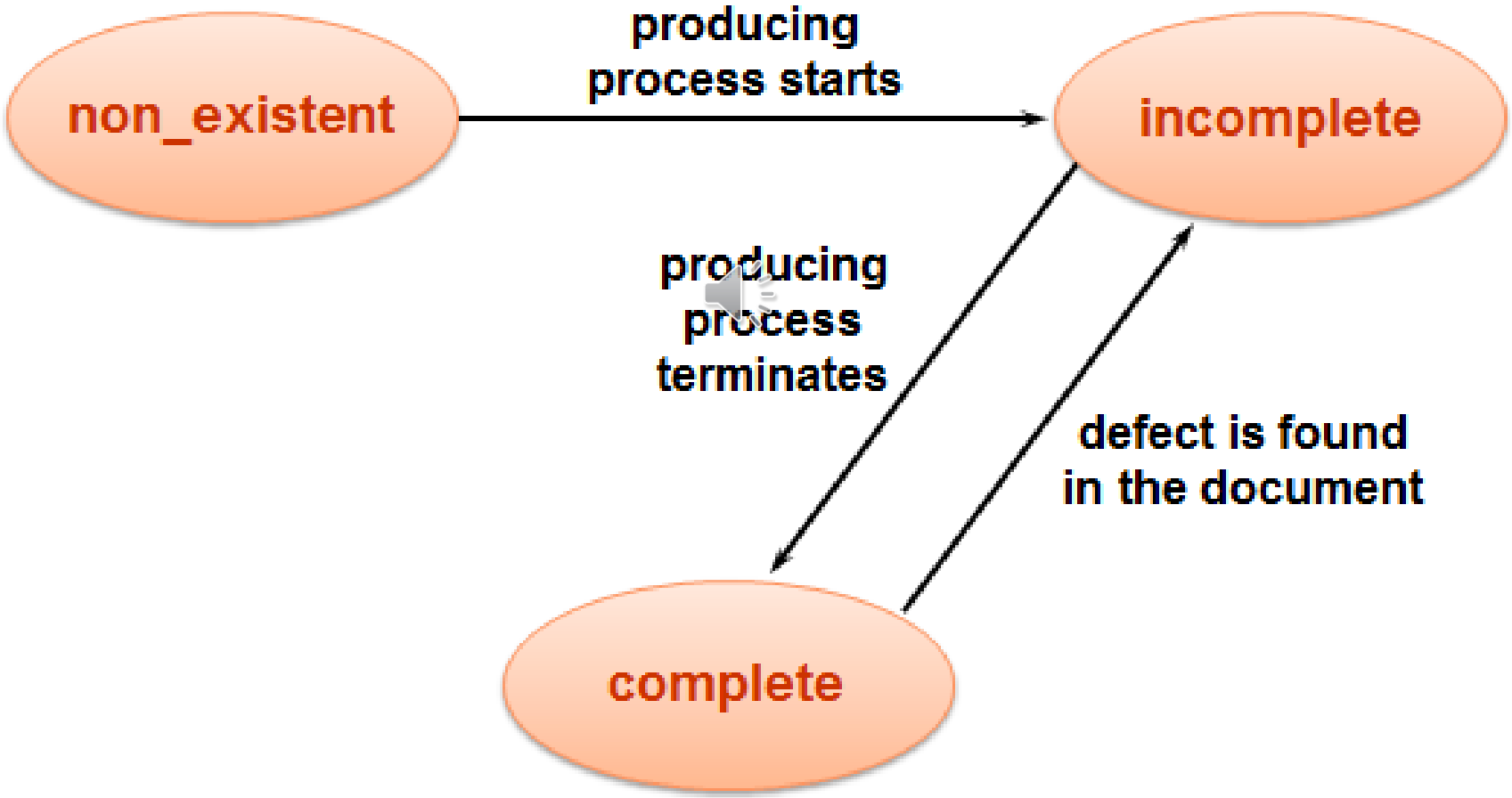



Products
Resources
Processes
Entry
Exit
Entry / Exit Criteria
consume
produce
produce/consume

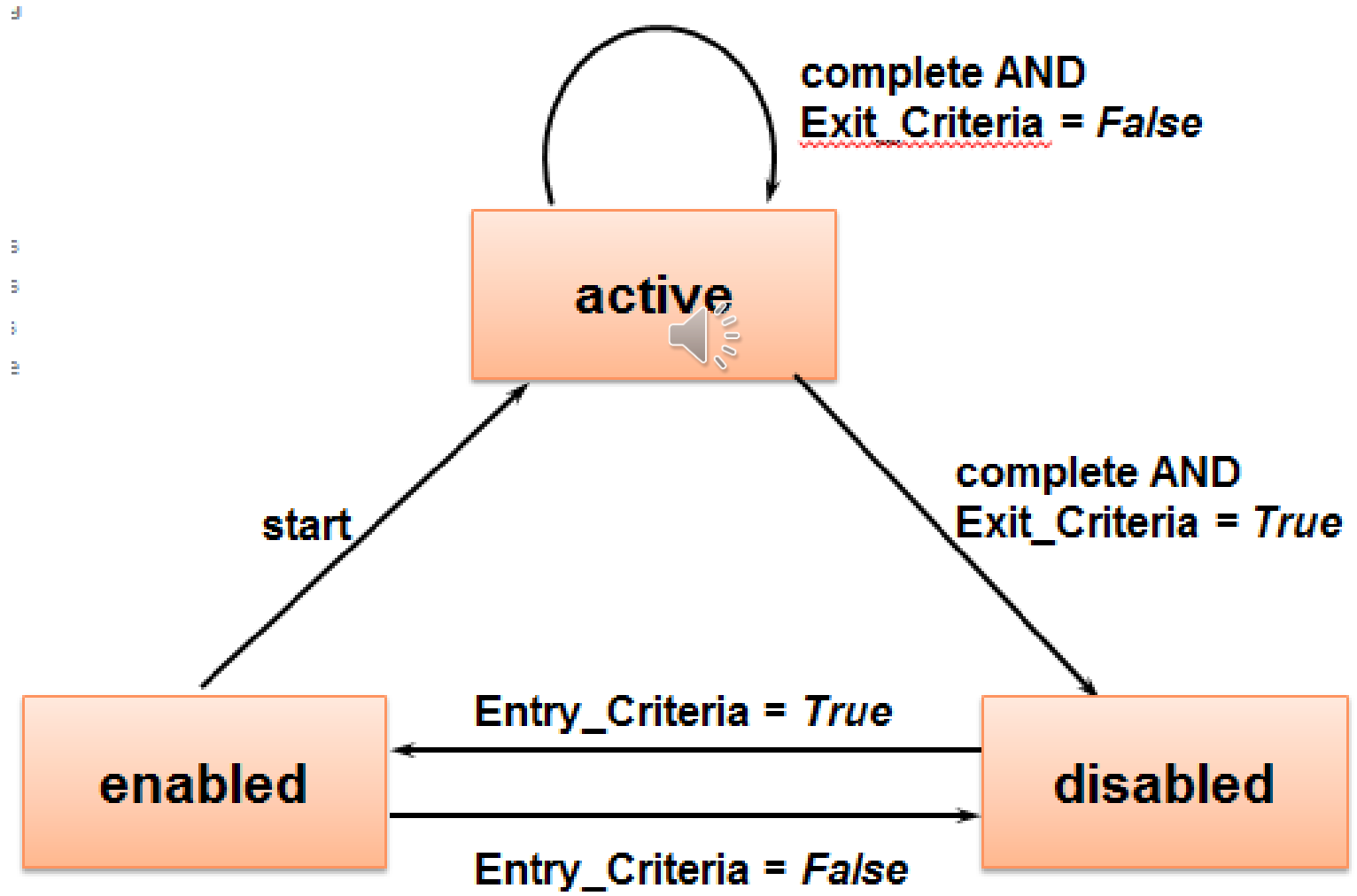
# Semantics

- Basic concepts
  - All processes are independent
  - Each product / each process has a state (defined by the values of all its attributes)
  - Each project plan has a state (defined by the states of all the products' processes)
  - State transitions are fired by external events:
    - Initiation of a process (start)
    - completion of a process (complete)
    - "enforced" state change (set)
- Instantiation of a project plan
  - An initial state is defined by
    - Binding formal interface parameters to actual objects
    - Specification of default parameters
- Execution (Process machine MVP-S)
  - State transitions are defined by state transition graphs in the attribute models
  - External events are ordered by the process machine

# Execution semantics: Transition Graph Product States



# Execution semantics: Transition Graph Process States

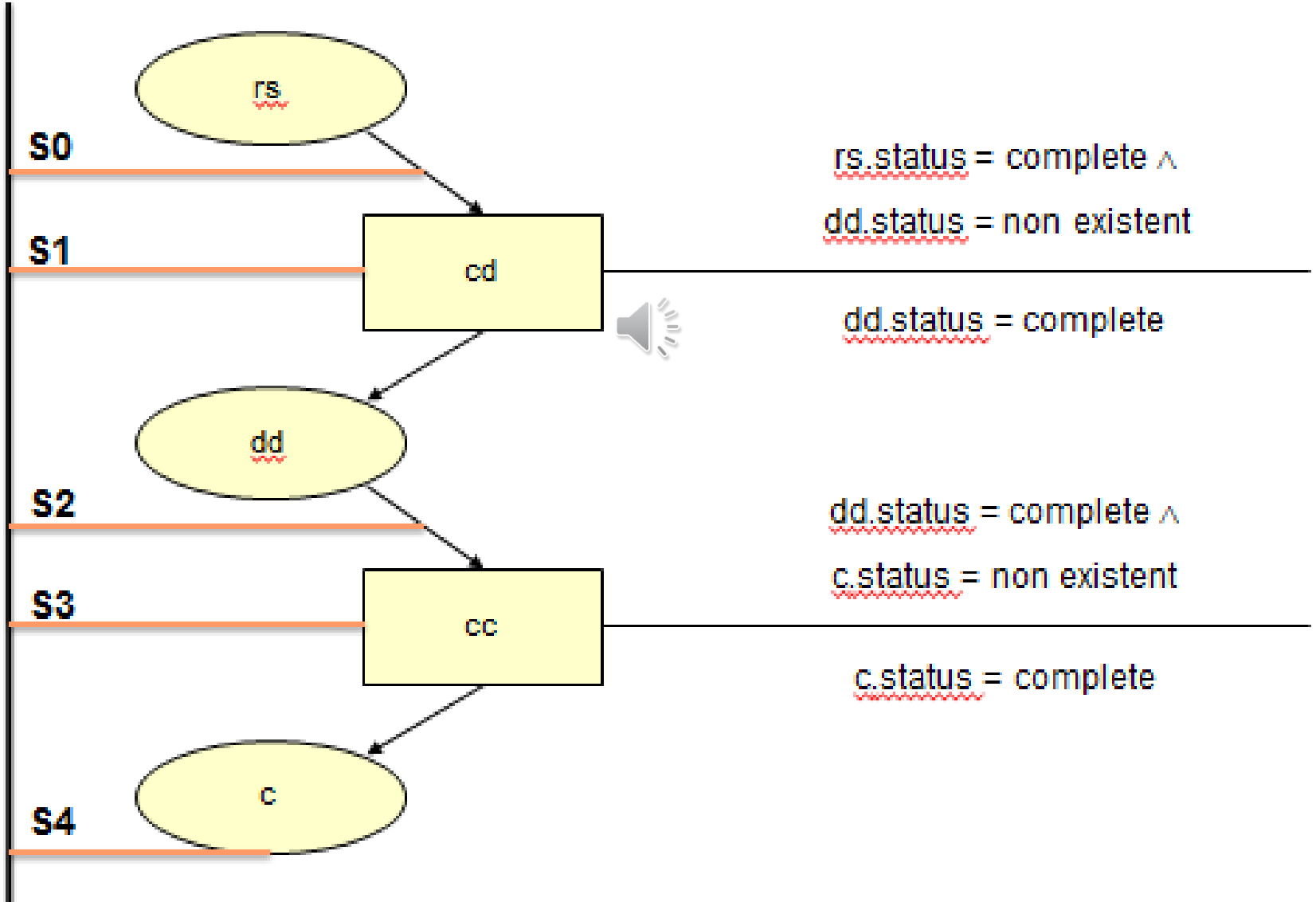


# Example

- Process models
  - cd : Create Design (“cd” is instance of Create Design)
  - cc : Create Code
- Product models
  - rs : Requirements Specification
  - dd : Design Document
  - c : Code
- Strict sequential process
- Control by pre- and post-conditions of the processes



# Process in MVP-L: Waterfall



# MVP-L – Execution Semantics: Transition Table for Waterfall

| | Status | S0 |
|---|---|---|
| Product | rs | complete |
| | dd | non existent |
| | c | non existent |
| Process | cd | enabled |
| | cc | disabled |



# MVP-L – Execution Semantics: Transition Table for Waterfall

| | Status | S0 | S1 | S2 | S3 | S4 |
|---|---|---|---|---|---|---|
| Product | rs | complete | complete | complete | complete | complete |
| | dd | non existent | incomplete | complete | complete | complete |
| | c | non existent | non existent | non existent | incomplete | complete |
| Process | cd | enabled | active | disabled | disabled | disabled |
| | cc | disabled | disabled | enabled | active | disabled |



**User Events:**

cd.start

cd.complete

cc.start

cc.complete